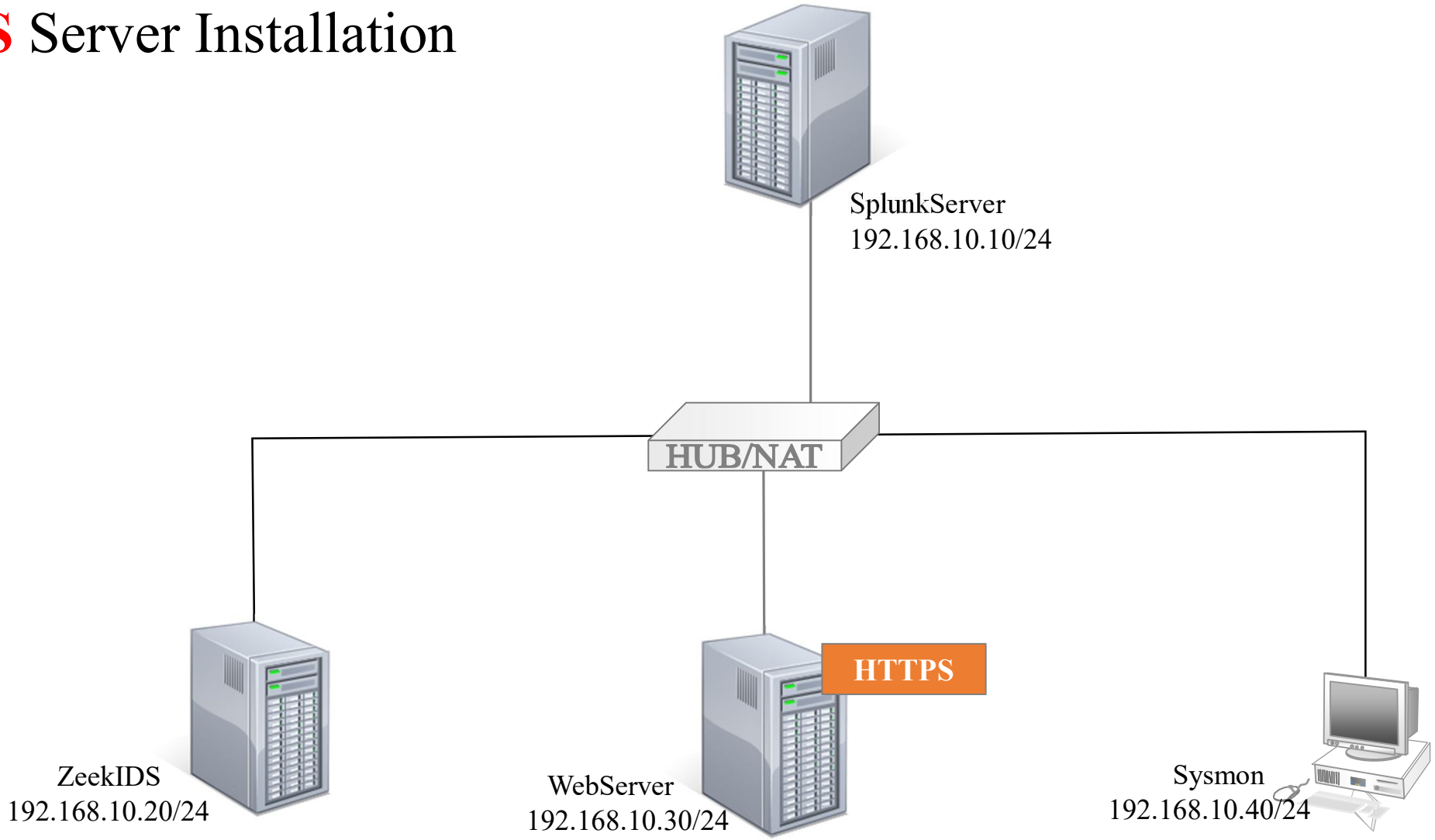
HTTPS Server Installation



[Rocky Linux 9] HTTP Server Installation

WebServer
192.168.10.30/24

① Apache package 설치

```
#rpm –qa httpd | grep httpd

#dnf install -y install httpd

#rpm –qa httpd

#rpm –qi httpd

#httpd -v
```

② index.html 구성

WebServer
192.168.10.30/24

#nano /var/www/html/index.html

```
<html>
<head><title>Open.NET Page</title></head>
<body>
<center><font size=5> This is MY Homepage</font><br><br>
</body>
</html>
```

```
[root@Rocky /]# cd /var/www/html
[root@Rocky html]# ls
hello.jpg  index.html
[root@Rocky html]# cat index.html
<html>
<head><title>Open.NET Page</title></head>
<body>
<center><font size=5> This is MY Homepage</font><br><br>
<img src="hello.jpg" width="300" height="300"></center>
</body>
</html>

[root@Rocky html]#
```

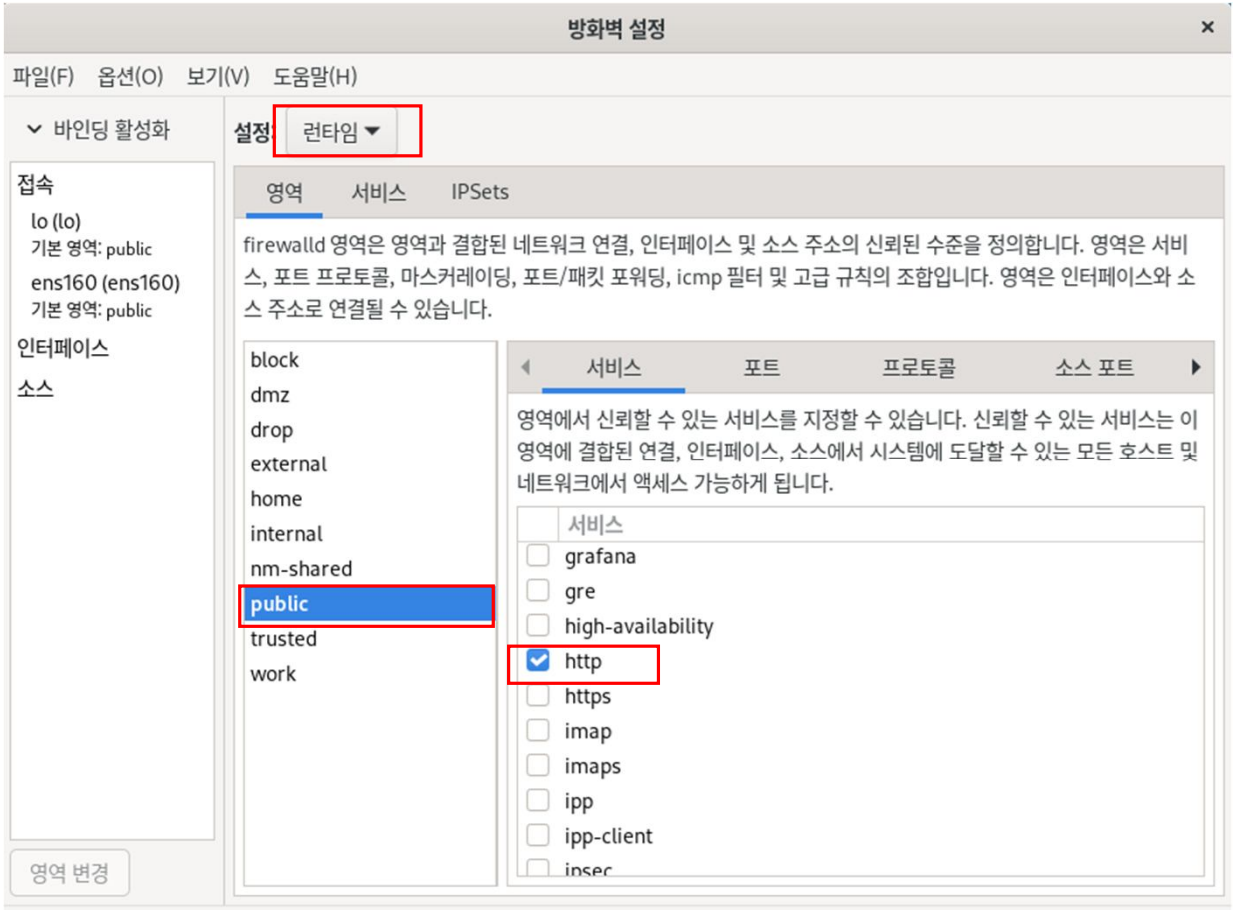
③ 서비스와 방화벽 활성화

#firewall-config

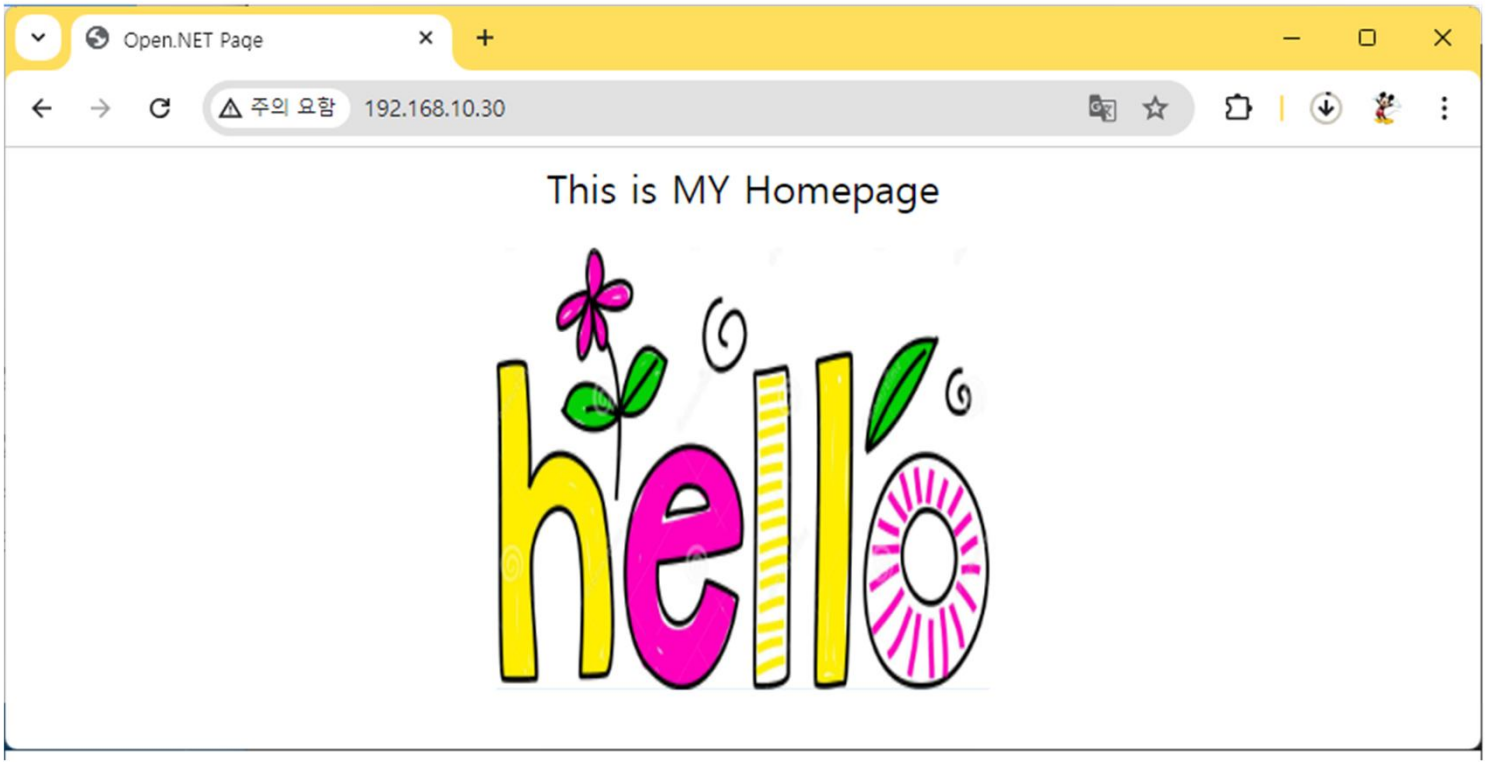
#systemctl start httpd

#systemctl enable httpd

#systemctl status



http://192.168.10.30

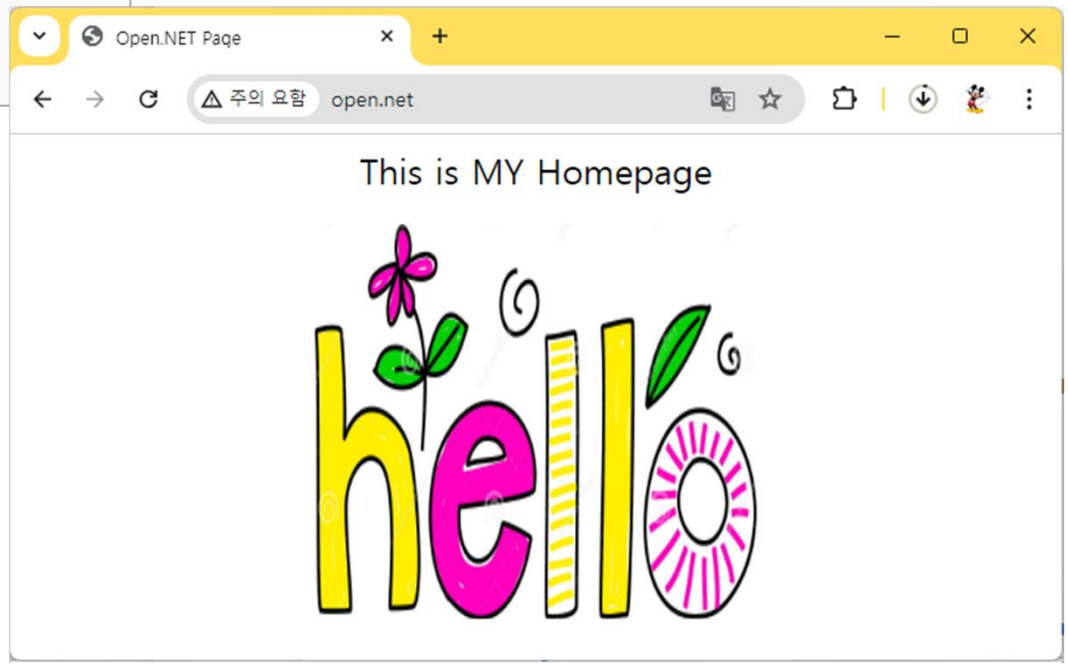


hosts

파일 편집 보기

```
# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost

192.168.10.30  www.open.net
```

http://www.open.net



- 환경 설정 파일

#cd /etc/httpd/conf

```
[root@rsyslog-client httpd]# ls
conf  conf.d  conf.modules.d  logs  modules  run  state
[root@rsyslog-client httpd]# cd conf
[root@rsyslog-client conf]# ls
httpd.conf  magic
[root@rsyslog-client conf]# gedit httpd.conf
[root@rsyslog-client conf]# pwd
/etc/httpd/conf
[root@rsyslog-client conf]#
```

- 로그 파일

#cd /var/log/httpd

```
[root@rsyslog-client httpd]# pwd
/var/log/httpd
[root@rsyslog-client httpd]# ls -l
합계 8
-rw-r--r--. 1 root root 1005  8월  7 12:57 access_log
-rw-r--r--. 1 root root 1602  8월  7 12:49 error_log
[root@rsyslog-client httpd]#
```

[Rocky Linux 9] HTTPS Server Installation

① OpenSSL 설치

#dnf install –y openssl*

#rpm –qa | grep openssl

② mod_ssl 모듈 확인

#httpd –l

#ls /etc/httdp/modules/mod_ssl.so

<<mod_ssl.so 파일이 없는 경우>>

rpm –qa |grep mod_ssl

#dnf install -y mod_ssl*

#ls /etc/httdp/modules/mod_ssl.s

③ 개인키 생성

```
#cd /etc/pki/tls/certs
#openssl genrsa –out www.open.net.key 2048
```

④ CSR 생성

```
#openssl req –new –key www.open.net.key -out www.open.net.csr
```

#openssl req –new –key www.open.net.key -out www.open.net.csr

```
[root@Rocky certs]# openssl req -new -key www.open.net.key -out www.open.net.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:KR
State or Province Name (full name) []:Seoul
Locality Name (eg, city) [Default City]:Seoul
Organization Name (eg, company) [Default Company Ltd]:OpenSource Corp.
Organizational Unit Name (eg, section) []:Security&System
Common Name (eg, your name or your server's hostname) []:www.open.net
Email Address []:admin@eopen.net

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
[root@Rocky certs]#
```

⑤ 개인키와 CSR을 이용해 인증서 생성

#openssl x509 –in www.open.net.csr –out www.open.net.crt –req –signkey www.open.net.key –days 365

#ls –l www.open.net*

```
[root@Rocky certs]# openssl x509 -in www.open.net.csr -out www.open.net.crt -req -signkey www.open.net.key
-days 365
Certificate request self-signature ok
subject=C = KR, ST = Seoul, L = Seoul, O = OpenSource Corp., OU = Security&System, CN = www.open.net, email
Address = admin@eopen.net
[root@Rocky certs]# ls -l www.open.net*
-rw-r--r--. 1 root root 1354  6월  30 15:55 www.open.net.crt
-rw-r--r--. 1 root root 1074  6월  30 15:52 www.open.net.csr
-rw-------. 1 root root 1704  6월  30 15:50 www.open.net.key
```

⑥ ssl.conf 파일에 인증서 경로 설정

#nano /etc/httpd/conf.d/ssl.conf

SSLCertificateFile /etc/pki/tls/certs/www.open.net.crt

....

SSLCertificateKeyFile /etc/pki/tls/certs/www.open.net.key

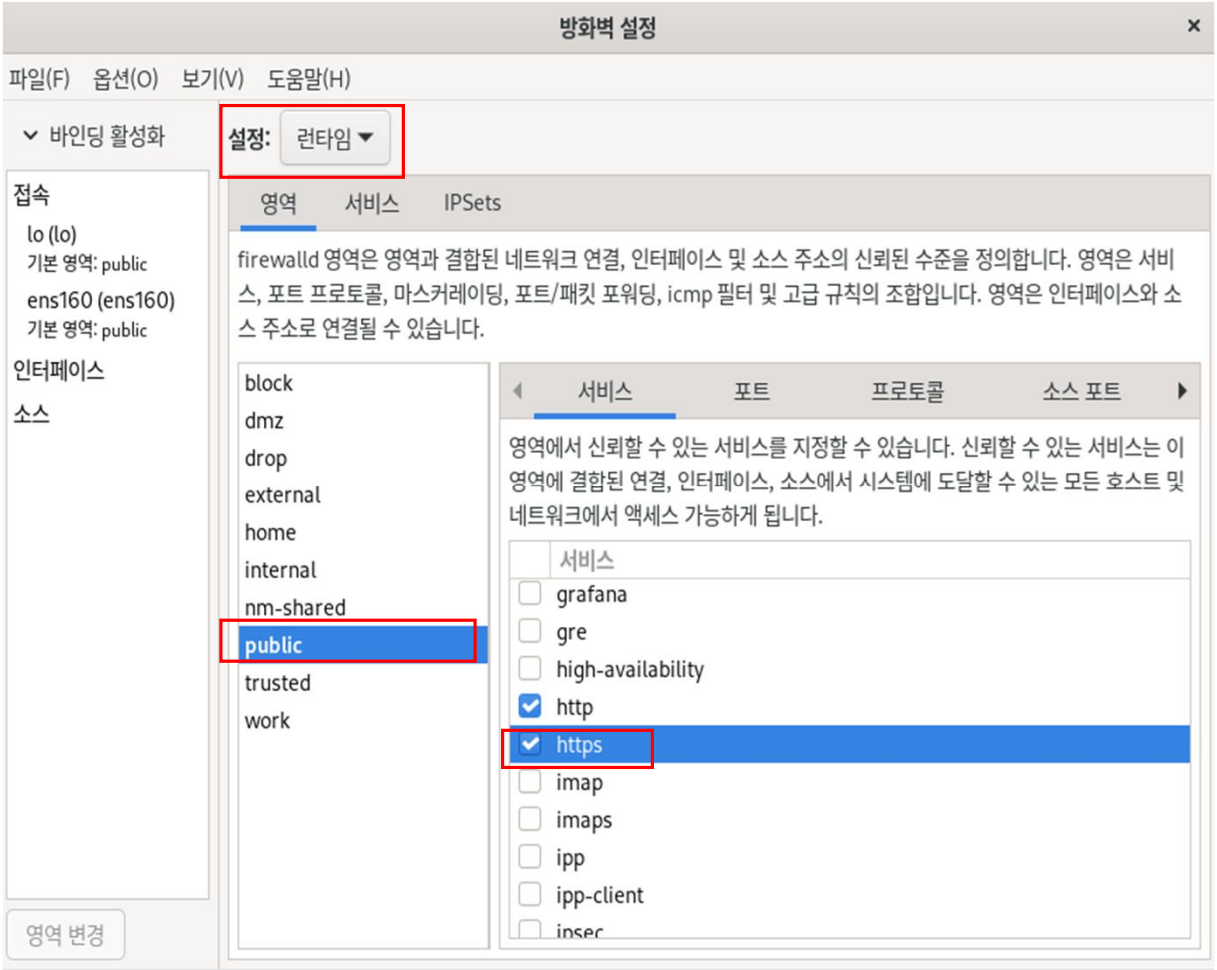
```
  GNU nano 5.6.1                                          ssl.conf
#   require an ECC certificate which can also be configured in
#   parallel.
SSLCertificateFile /etc/pki/tls/certs/www.open.net.crt

#   Server Private Key:
#   If the key is not combined with the certificate, use this
#   directive to point at the key file.  Keep in mind that if
#   you've both a RSA and a DSA private key you can configure
#   both in parallel (to also allow the use of DSA ciphers, etc.)
#   ECC keys, when in use, can also be configured in parallel
SSLCertificateKeyFile /etc/pki/tls/certs/www.open.net.key

#   Server Certificate Chain:
```

⑦ 서비스와 방화벽 활성화

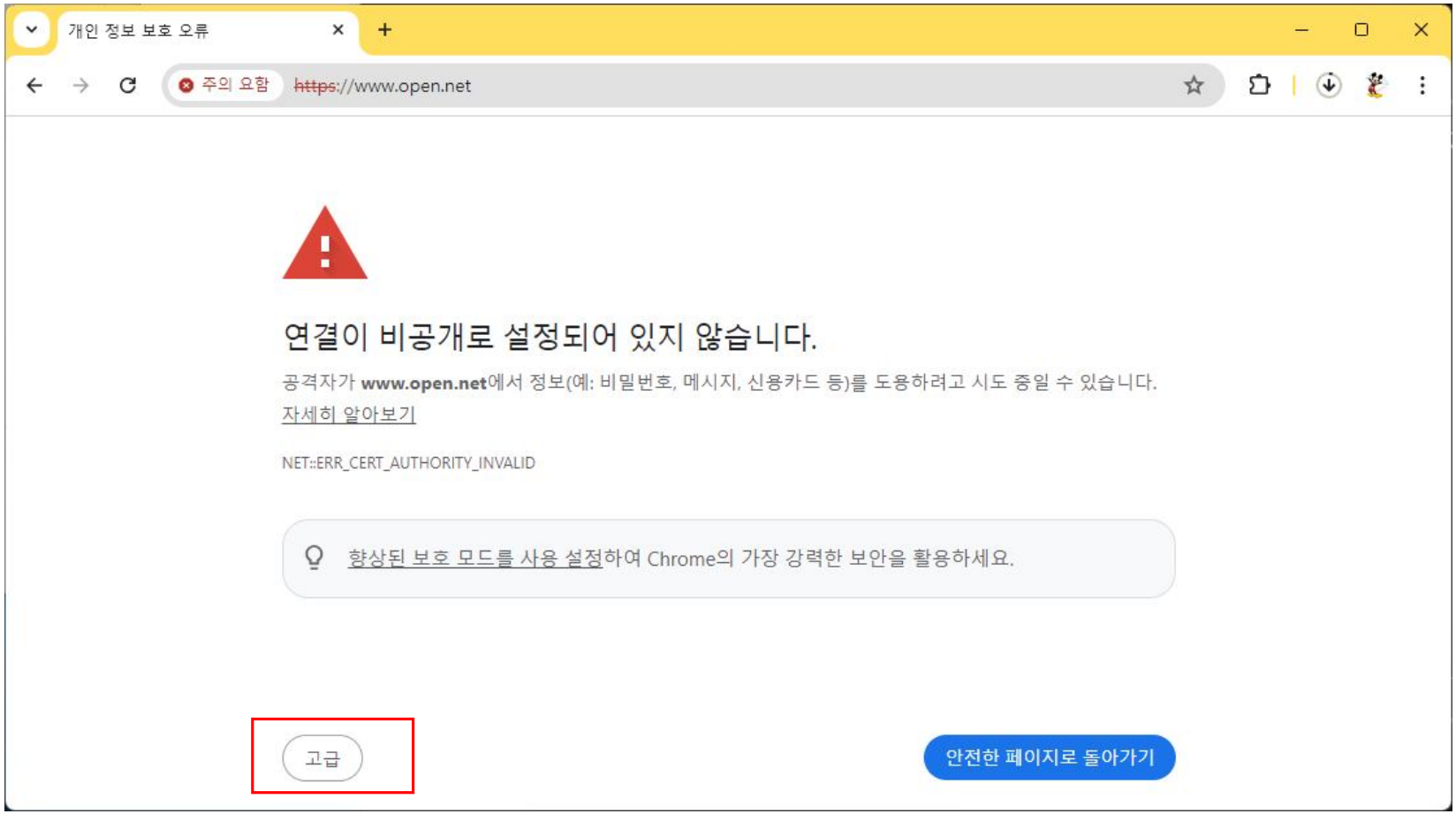
#firewall-config

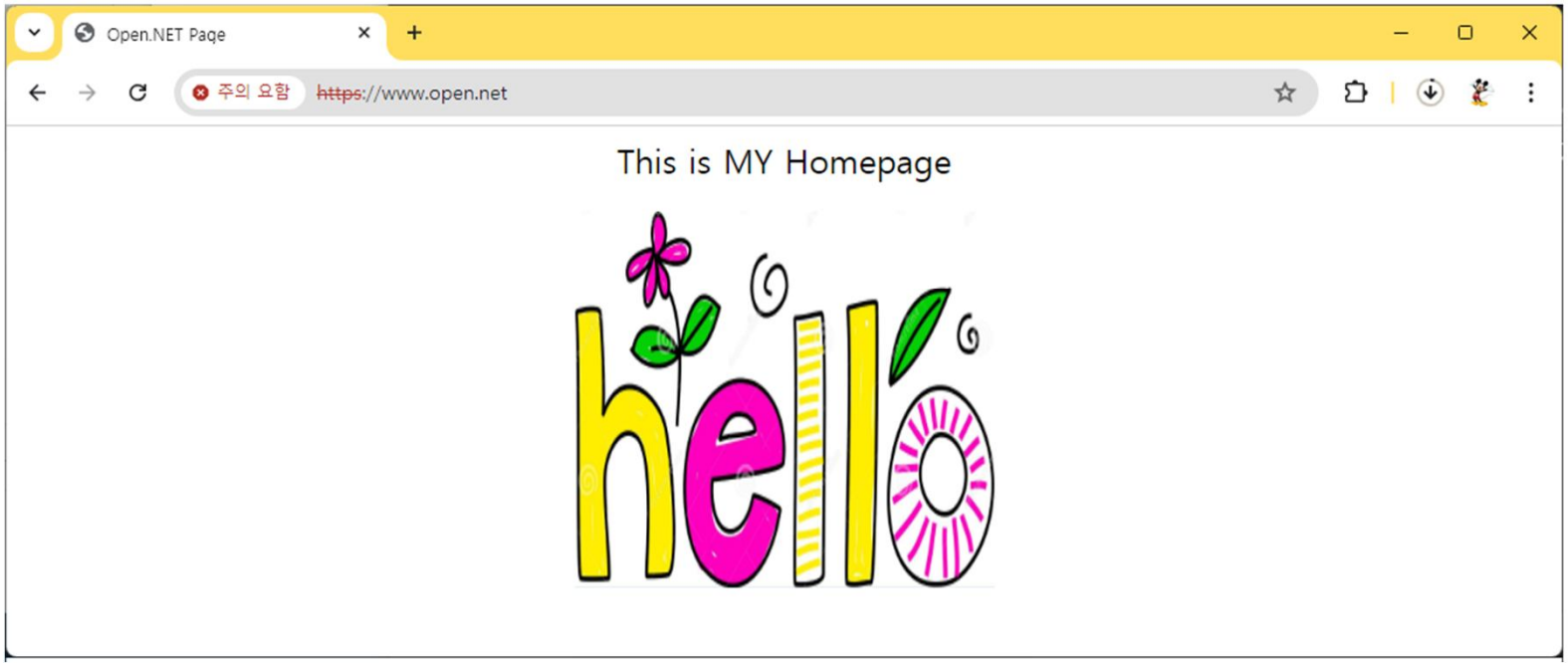
#systemctl restart httpd

#systemctl status httpd

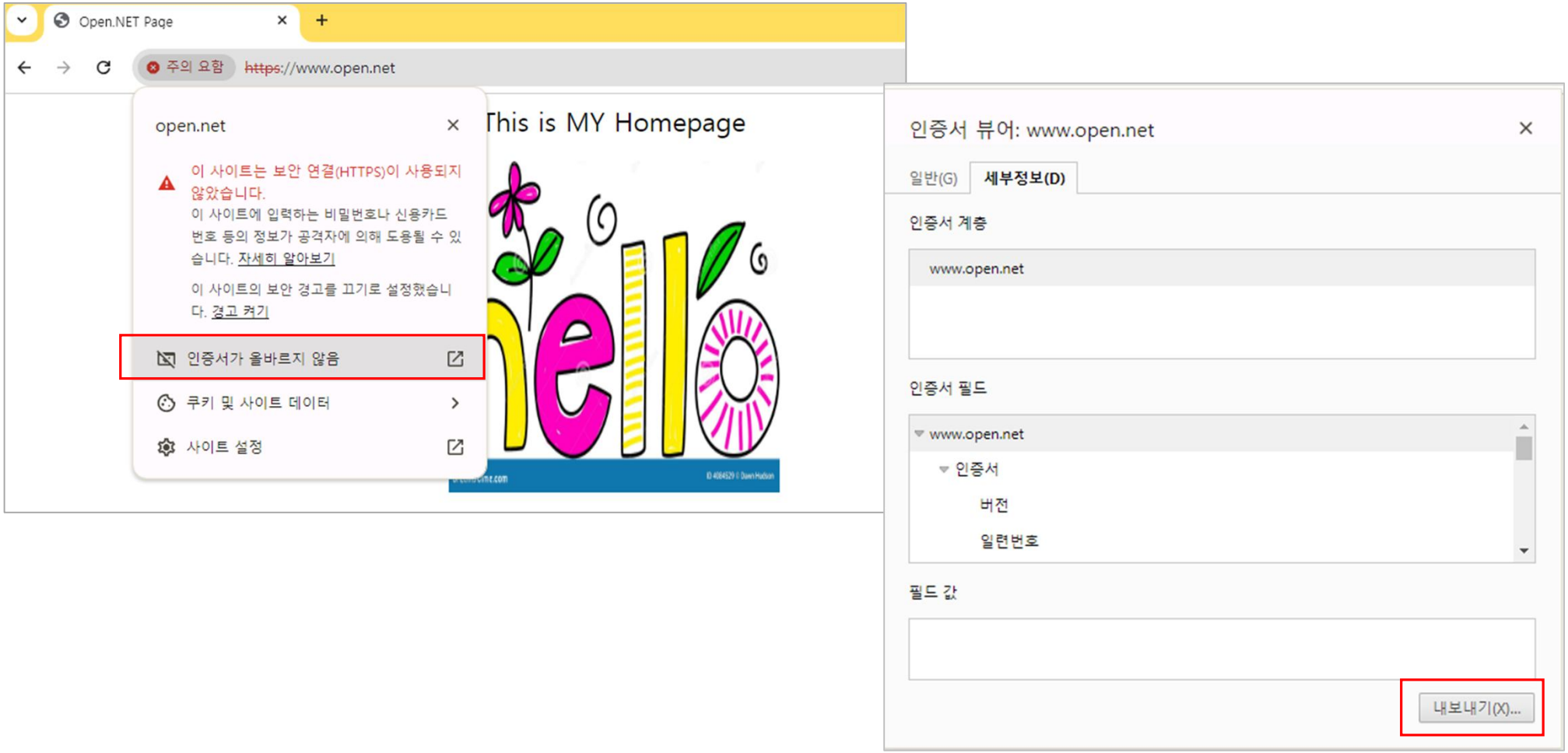


https://www.open.net

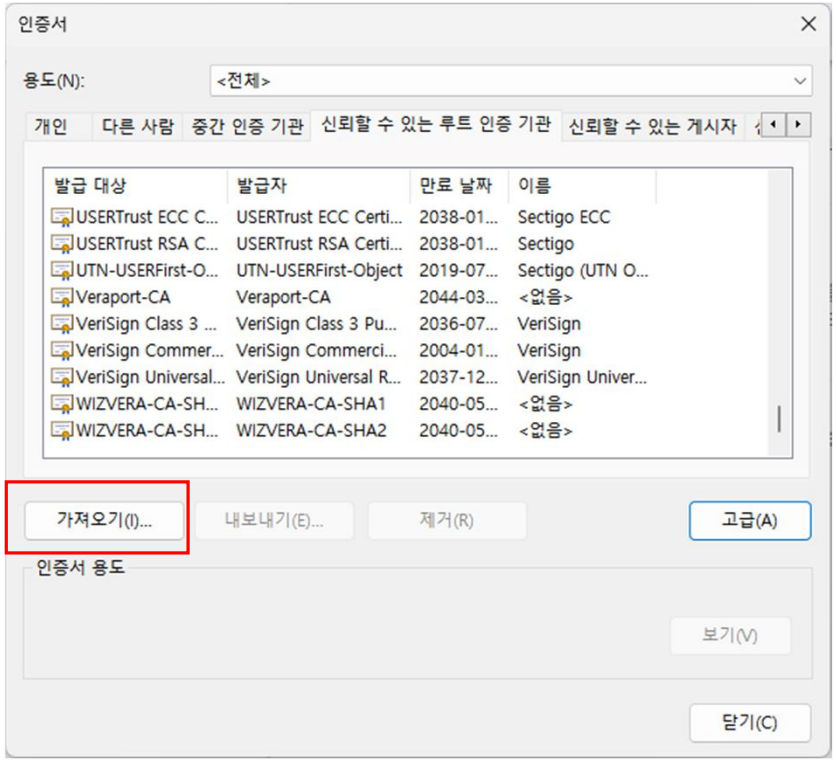


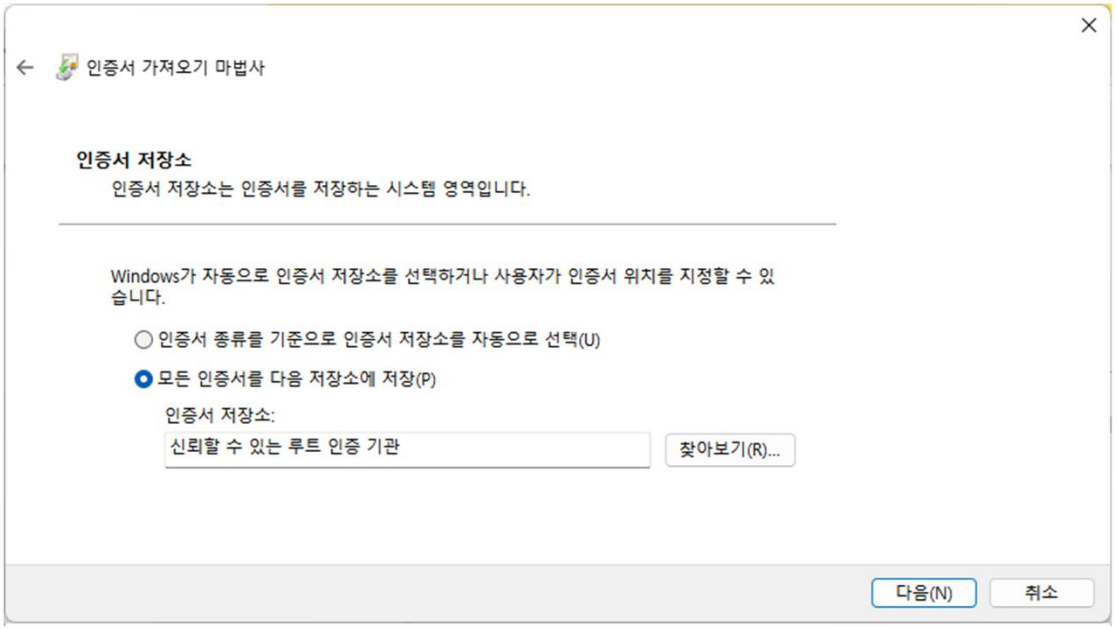

Open.NET Page
주의 요함
https://www.open.net
This is MY Homepage
hello

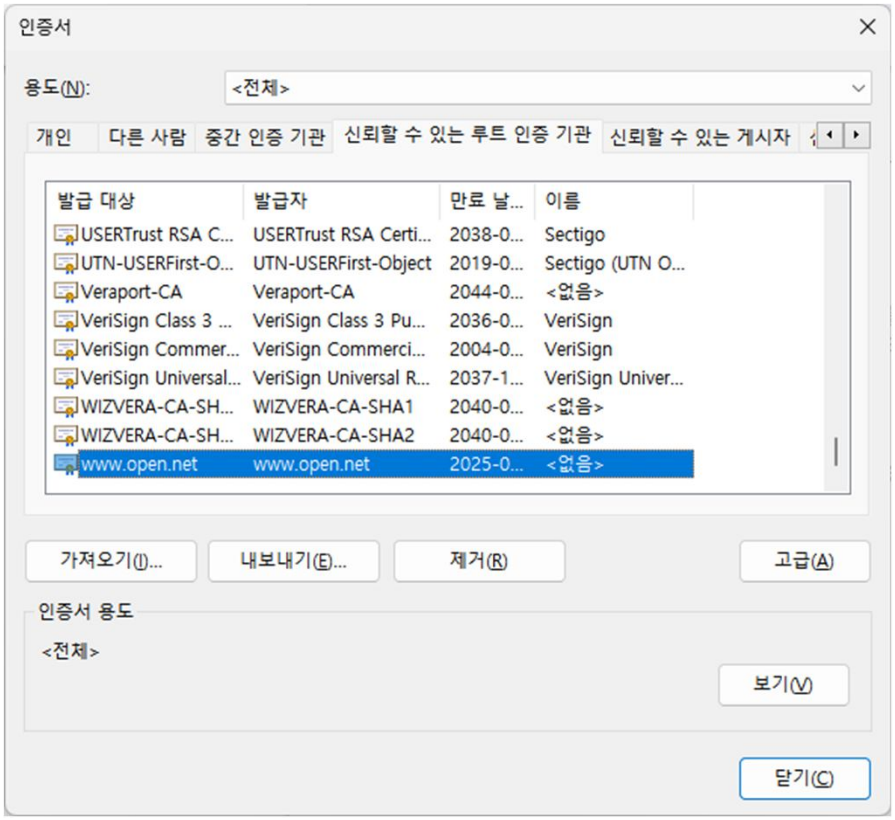
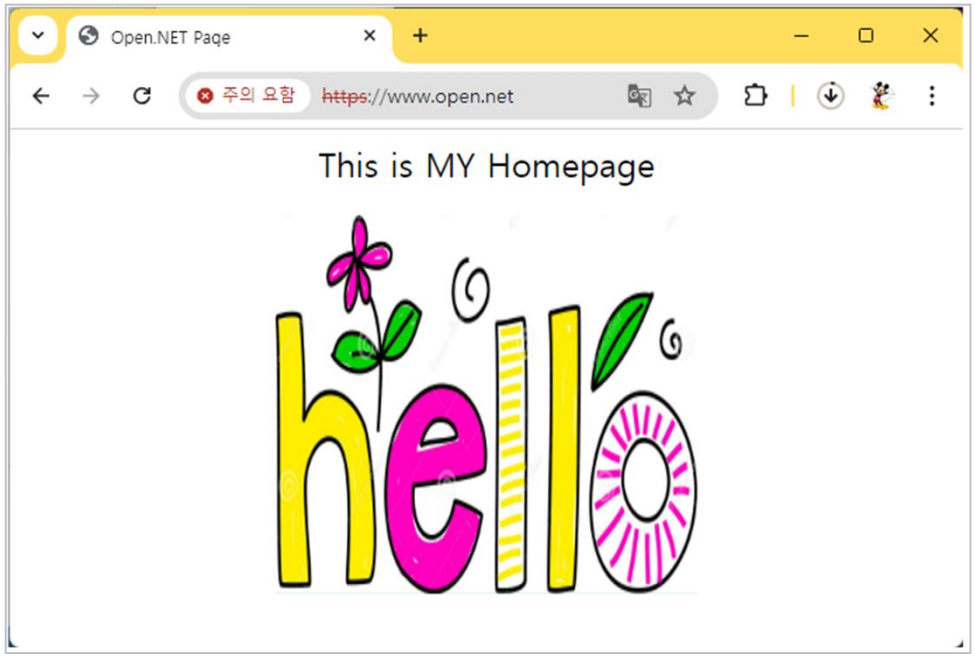
⑧ 인증서 확인 및 내보내기



⑨ ‘신뢰할 수 있는 루트 인증기관 ' 에 인증서 설치







https://www.open.net